

Experiment No: 1**Experiment Title:** Microsoft Threat Modeling using STRIDE**AIM**

To create a threat model for a simple shopping application and identify security threats using STRIDE.

REQUIREMENTS

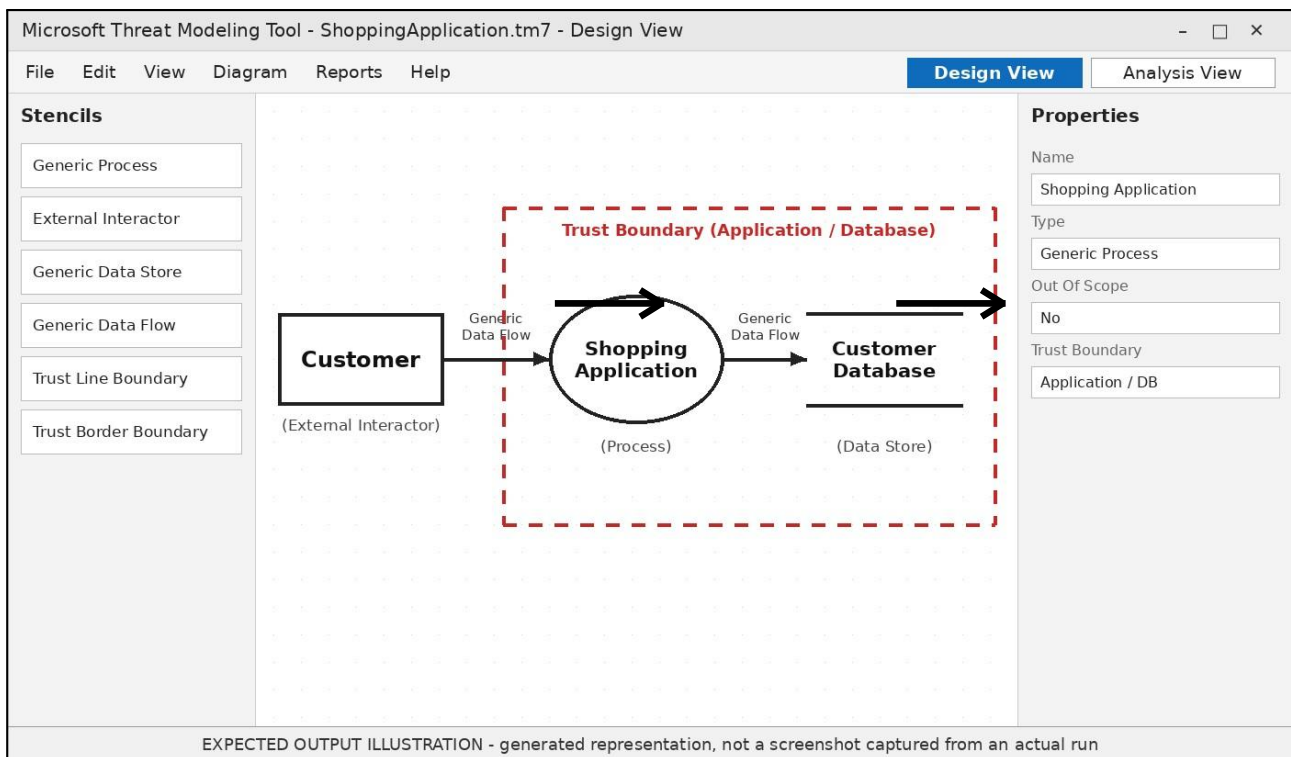
- Microsoft Threat Modeling Tool
- Windows system
- Sample application/data-flow diagram

PROCEDURE

1. Open the Threat Modeling Tool and create a new model.
2. Add Customer, Shopping Application and Customer Database.
3. Connect the elements with data-flow connectors.
4. Add a Trust Boundary.
5. Run analysis and review the generated STRIDE threats.
6. Record suitable mitigations such as authentication, encryption, logging and least privilege.

STRIDE THREAT CATEGORIES

STRIDE category	Meaning (standard definition)
Spoofing	Pretending to be another user or component
Tampering	Unauthorized modification of data or code
Repudiation	Denying an action without the ability to prove otherwise
Information Disclosure	Exposure of information to unauthorized parties
Denial of Service	Making a service unavailable
Elevation of Privilege	Gaining capabilities without proper authorization

**Figure 1: Data Flow Diagram of the Shopping Application**

Expected Output: Design View with Customer (External Interactor), Shopping Application (Process) and Customer Database (Data Store) joined by Generic Data Flow connectors, with a Trust Boundary around the Application / Database area.

Microsoft Threat Modeling Tool - ShoppingApplication.tm7 - Analysis View

File Edit View Diagram Reports Help

Design View **Analysis View**

Threat List - threats generated using the STRIDE model

Id	Threat	Category	Priority	Status
1	Spoofing of the Customer External Entity	Spoofing	High	Not Started
2	Potential Data Tampering in the Data Flow	Tampering	High	Not Started
3	Potential Data Repudiation by Shopping Application	Repudiation	Medium	Not Started
4	Weak Access Control for a Resource	Information Disclosure	High	Not Started
5	Potential Process Crash or Stop	Denial of Service	Medium	Not Started
6	Elevation Using Impersonation	Elevation of Privilege	High	Not Started

Recorded mitigations (per practical): authentication, encryption, logging, least privilege

Analysis status: threat analysis completed

EXPECTED OUTPUT ILLUSTRATION - generated representation, not a screenshot captured from an actual run

Figure 2: Generated STRIDE Threat List

Expected Output: Analysis View listing a generated threat for each STRIDE category (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) with Category, Priority and Status columns. Threat titles are illustrative; your tool will generate its own.

RESULT:

Using the Microsoft Threat Modeling Tool, a data-flow model of a simple shopping application was analysed and threats were categorized under the STRIDE model, with mitigations such as authentication, encryption, logging and least privilege recorded.